

Day 40 - SNMP

Sunday, September 6, 2026

10:46 AM

SNMP Info

SNMP General Information

Simple Network Management Protocol (SNMP) is an *industry-standard* framework and protocol that can be used to monitor the status of devices, make configuration changes and etc.

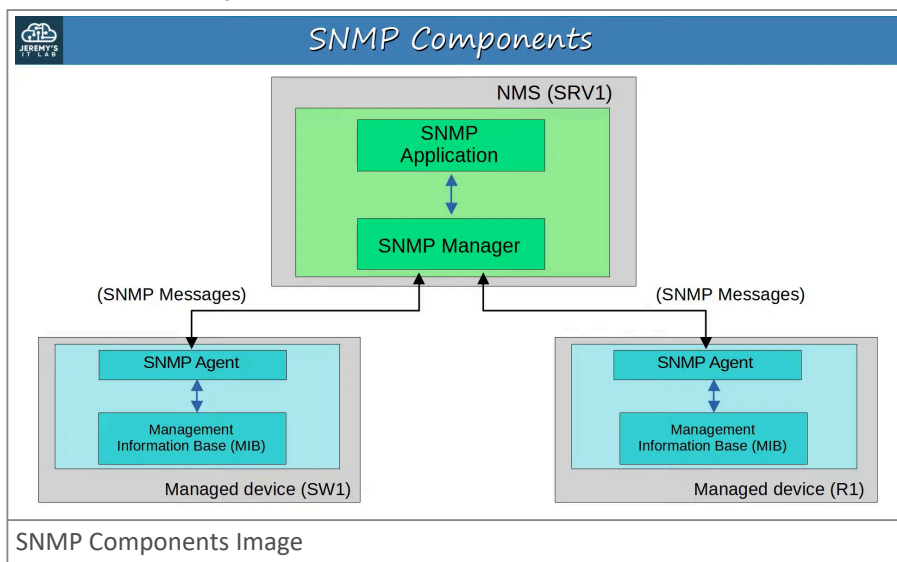
- **SNMP** agents (clients) run on port **161** while **SNMP** managers (servers) run on port **162**.
- There are two main types of devices in **SNMP**
 1. **Managed Devices**
 - These are the devices being managed using **SNMP**.
 - (Ex. network devices like routers and switches).
 2. **Network Management Station (NMS)**
 - The device/devices managing the *managed devices*.
 - This is the **SNMP** 'server'.

SNMP Main Operations

There are three main operations used in SNMP

1. *Managed devices* can notify the **NMS** of events.
2. The **NMS** can ask the *managed devices* for information about their current status.
3. The **NMS** can tell the *managed devices* to change aspects of their configuration.

SNMP Components



1. NMS

The **NMS** has two SNMP components, being:

1. **SNMP Manager**
 - It is the **SNMP** software on the **NMS** that interacts with the managed devices.
 - It receives notifications, sends requests for information, sends configuration changes, and etc.
2. **SNMP Application**
 - Provides an interface for the network admin to interact with.
 - Displays alerts, statistics, charts and etc.

2. Managed Devices

The **Managed devices** run two SNMP components, being:

1. SNMP Agent

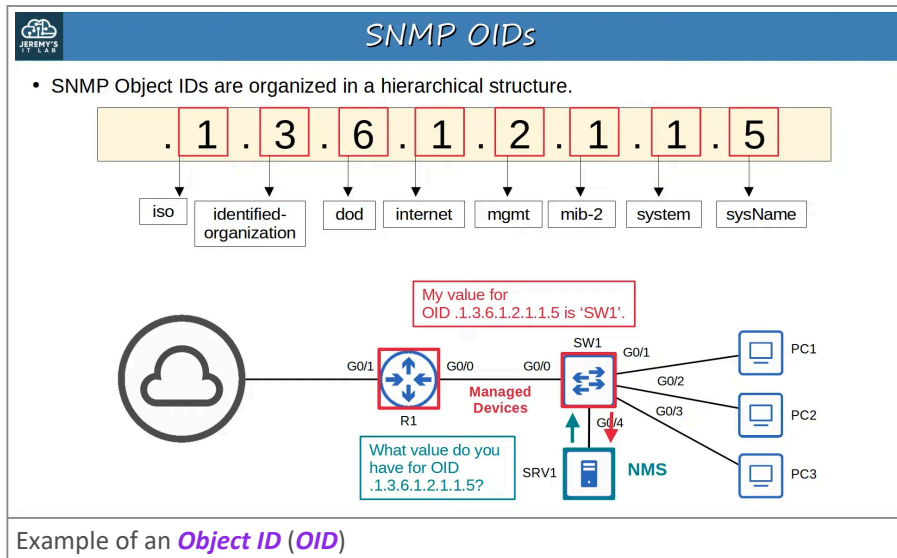
- It is the **SNMP** software running on the **managed devices** that interacts with the **SNMP** manager on the **NMS**.

2. SNMP Management Information Base (MIB)

- It is the structure that contains the **variables** that are managed by **SNMP**
- Each **variable** is identified with an **Object ID (OID)**
- Example **variables**: Interface status, traffic throughput, CPU usage, temperature, and etc.

SNMP Object IDs (OIDs)

SNMP **Object IDs (OID)** are organized in a hierarchical structure



SNMP Versions

A lot of SNMP versions have been developed. The most used three versions are:

1. SNMPv1

- The original version of **SNMP**

2. SNMPv2c

- Allows the **NMS** to retrieve large amounts of information in a single request, so it is more efficient (**GetBulk message**)
- The 'c' in the name refers to **community strings** which are used as passwords

3. SNMPv3

- Secure version of **SNMP** that supports **encryption** and **authentication**.

SNMP Messages

SNMP Message Types Table

Message Class	Description	Messages
Read	Messages sent by the NMS to read information from the managed devices .	Get GetNext GetBulk
Write	Messages sent by the NMS to change information on the managed devices .	Set
Notification	Messages sent by the managed devices to alert the NMS of a particular event.	Trap Inform
Response	Messages sent in response to a previous message/request.	Response

1. SNMP Read Messages

1. Get

- A request sent from the **manager** to the **agent** to retrieve the value of a **variable (OID)**, or multiple **variables**.
- The agent will send a **Response** message with the current value of each variable

2. GetNext

- A request sent from the **manager** to the **agent** to discover the available variables in the **Management Information Base (MIB)**

3. GetBulk ^snmp-messages-getbulk

- A more efficient version of the **GetNext** message. (Introduced in [SNMPv2](#))

2. SNMP Write Messages

1. Set

- A request sent from the **manager** to the **agent** to change the value of one or more **variables**.
- The **agent** will send a **Response** message with the new updated values.

3. SNMP Notification Messages

1. Trap

- A notification sent from the **agent** to the **manager**.
- The **manager** does not send a **Response** message to acknowledge that it has received the **Trap**, so these messages are considered 'unreliable'.

2. Inform

- A notification message that is acknowledged with a **Response** message

4. SNMP Response Messages

1. Response

- Sent as a response to a variety of messages, like the ones previously mentioned.

SNMP Configuration on Cisco IOS

- **snmp-server contact CONTACT-INFO**
 - Sets the **SNMP** server contact information
- **snmp-server location LOCATION-INFO**
 - Sets the **SNMP** server location information
- **snmp-server community PASSWORD ro**
 - Sets the **community password** for read-only operations
- **snmp-server community PASSWORD rw**
 - Sets the **community password** for read-write operations
- **snmp-server host IP-ADDRESS version 2c COMMUNITY-STRING**
 - Specifies the **NMS**, version and **community string**
- **snmp-server enable traps TRAP-TYPES**
 - Enables specific **trap** (notification) types